

Dongwook Kwon

AI Engineer · Intelligent Agent Technology

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RESEARCH INTERESTS

Anomaly detection in time-series data (ECG, cyber-physical systems); agentic AI and multi-agent LLM systems; graph neural networks for biomedical and industrial signals; verification and evaluation of AI systems.

EDUCATION

M.S. in Computer Engineering

Kwangwoon University

Mar 2024 – Feb 2026

Seoul, Korea

GPA 4.33 / 4.50 (98.1%) · Bio-Computing & Machine Learning Laboratory.

Graduate Research Program in Mechanical & Industrial Engineering

University of Toronto

Jan 2025 – Jul 2025

Toronto, Canada

GPA 3.68 / 4.00.

B.S. in Electronics & Communications Engineering

Kwangwoon University

Mar 2019 – Feb 2024

Seoul, Korea

GPA 3.41 / 4.50 (88.1%).

PUBLICATIONS

Journal Articles

1. Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park. “Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise.” *Electronics*, vol. 14, no. 5, p. 890, 2025.
2. **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park. “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments.” Under review at *Data Mining and Knowledge Discovery* (Springer Nature), 2025.

Conference Papers

1. R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma. “CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades.” In *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) – Industry Track (Poster)*, 2026. [†]Equal contribution.
2. J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma. “GEMMAS: Graph-based Evaluation Metrics for Multi-Agent Systems.” In *Proc. EMNLP 2025 Industry Track (Oral)*, 2025. [†]Equal contribution.
3. **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park. “Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers.” In *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. – *Best Poster Award, Silver*.

4. **D. Kwon**, Y. Kang, and C. Park. “Enhancing medical device security with GNN-GRU anomaly detection model.” In *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. – *Outstanding Paper Award*.

PATENTS

1. “Method for Implantable Medical Device to Detect Anomaly in Real Time,” KR Patent No. 10-2961984, issued Apr 30, 2026.
2. “GPT API-Based Network Anomaly Detection,” KR Patent No. 10-2848814, issued Aug 18, 2025.

RESEARCH EXPERIENCE

AI Engineer, Intelligent Agent Technology Team

Suresoft Technologies

Dec 2025 – Present

Seongnam, Korea

Developing AI validation systems based on agentic architectures; designing autonomous and collaborative AI agents to verify and evaluate AI models; building scalable, reliable agentic verification frameworks.

Research Assistant

Bio-Computing & Machine Learning Laboratory, Kwangwoon University

Aug 2021 – Dec 2025

Seoul, Korea

Researched deep-learning algorithms for anomaly detection in time-series data including ECG and cyber-physical systems. Contributed to the development of an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

Visiting Researcher

LG Electronics Canada (with University of Toronto)

Jan 2025 – Jul 2025

Toronto, Canada

Developed an LLM-driven assessment framework for evaluating performance of agentic AI systems.

Summer Researcher

Qualcomm Institute, UC San Diego – AI Development Project

Jul 2022 – Aug 2022

San Diego, CA, USA

Participated in the Personality-Based Drug Addiction Prediction System project; attended seminars on deep learning and machine learning.

HONORS AND AWARDS

2025 **Silver Award**, 23rd Embedded Software Competition (KIISE), South Korea.

2025 **Outstanding Student Paper Award**, IEIE Conference.

2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC).

2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical & Biological Engineering (KOSOMBE).

2022 **Bronze Award**, Hanium ICT Mentoring Competition, FKII.

2022 **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX).

2022 **Achievement Award**, Qualcomm Institute AI Development Project, UC San Diego.

2021 **Gold Award (Ministerial)**, Smart Maritime Logistics Project, Minister of Oceans and Fisheries.

2021 **Grand Prize**, MY Capstone Design Competition, Kwangwoon University.

SELECTED FUNDED PROJECTS

AI-based Intrusion Detection and Attack Packet Crafting for APR1400

Sejong University (industry-academia)

Jul 2022 – Present

Seoul, Korea

AI defense system protecting nuclear power plant control networks from cyber attacks.

Active Kill-Switch and Biomarker-Based Defense for IoT Medical Devices

Kookmin University (industry-academia)

Sep 2021 – Dec 2022

Seoul, Korea

Security system for cardiac implantable devices using anomaly detection.

Quadruped Robot Using Vision SLAM with Depth Camera

Ministry of Science and ICT

Apr 2022 – Nov 2022

Seoul, Korea

Autonomous navigation algorithm with SLAM and depth-camera perception.

Smart Logistics System Using Computer Vision

Ministry of Oceans and Fisheries

Apr 2021 – Nov 2021

Seoul, Korea

Logistics detection system with location tracking and stack-layer identification.

LEADERSHIP AND SERVICE

- **President**, Kwangwoon International Student Association – intercultural programs and engagement initiatives.
- **Director of Filming & Editing**, Kwangwoon Broadcasting Center (KWBC) – media production oversight.
- **Class Representative**, Department of Electronics and Communications Engineering.
- **Executive Member**, Amateur Radio Club – workshops and technical exhibitions.